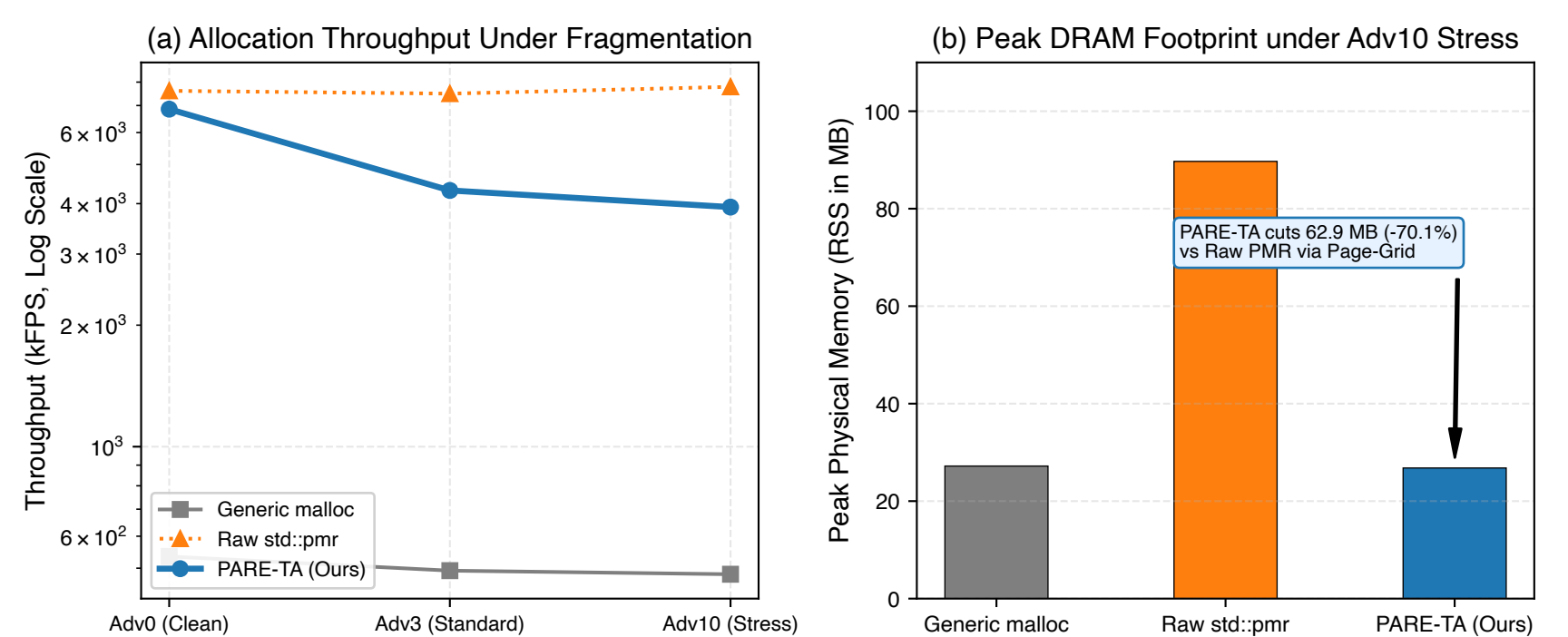
(a) Allocation Throughput Under Fragmentation
Throughput (kFPS, Log Scale)
6 × 10^3
4 × 10^3
3 × 10^3
2 × 10^3
10^3
6 × 10^2
Generic malloc
Raw std::pmr
PARE-TA (Ours)
Adv0 (Clean)
Adv3 (Standard)
Adv10 (Stress)
(b) Peak DRAM Footprint under Adv10 Stress
Peak Physical Memory (RSS in MB)
100
80
60
40
20
0
PARE-TA cuts 62.9 MB (-70.1%)
vs Raw PMR via Page-Grid
Generic malloc
Raw std::pmr
PARE-TA (Ours)